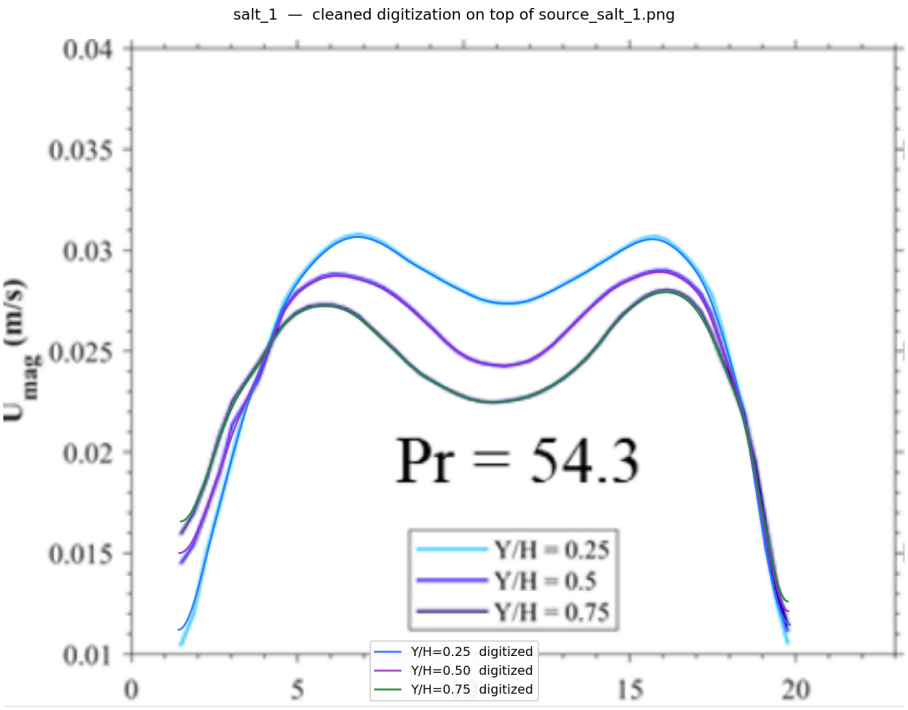


Figure 4.4: Digitized Velocity Profile of Salt Test 1 Overlaid on Source Image

the profile, as the ends of the profiles have the greatest error associated with them.

From the results in Table 4.6, the digitization uncertainty assigned to the water velocity profiles is rounded up from the tabulated L_2 percent error of 1.36% to the assigned value of

$$u_{dig} = 1.5\% \tag{4.11}$$

and is intended to compound upon the experimentally reported PIV uncertainty for each water case for calculations dependent on the full-field velocity data, such as the calculated value for mass flow rate in each experiment. This will be further discussed in Chapter 6

4.5.4 Data Uncertainty Summary

To conclude this chapter, we can summarize each contribution to u_D in the table below. The entry u_{PIV} is the uncertainty prescribed to each of the full-field PIV datasets reported by Reis et al. in [9]. The comparisons performed in Chapter 6 are mapped to each experimental